Direction of vorticity and continuation for 3D Navier–Stokes: the $\frac{1}{2}$-Hölder criterion on the maximal classical interval

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Abstract

Let u be the unique classical solution of the 3D incompressible Navier–Stokes equations on the maximal interval $[0, T^*)$, viscosity $\nu > 0$. Write $\omega = \nabla \times u$, $r = |\omega|$, $\xi = \omega/r$ on $\{r > 0\}$, and $S = \frac{1}{2}\big(\nabla u + (\nabla u)^\top\big)$. The two constants of the criterion are

$$\Omega = \Omega(u_0, \nu) > 0, \qquad C = C(u_0, \nu) < \infty.$$

If

$$|\xi(x,t) - \xi(y,t)| \le C\,|x-y|^{1/2}$$

whenever $r(x,t), r(y,t) \ge \Omega$ and $t < T^*$, then $T^* = \infty$ and u is globally smooth. The direction ξ is never evaluated on $\{r = 0\}$. No additional geometric term is added to the PDE. The implication is the Beirão da Veiga–Berselli $\frac{1}{2}$-Hölder form of the Constantin–Fefferman criterion. **Definition 7.1 is the only missing analytic ingredient; once it holds, the implication is unconditional.** It is not proved below. Energy does not force it. That is Theorem C.

MSC-style keywords: Navier–Stokes, vorticity direction, Beale–Kato–Majda, Constantin–Fefferman, Beirão da Veiga–Berselli.

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*No tether term is used. Degeneracy of $\xi = \omega/|\omega|$ is respected. Definition 7.1 $\Rightarrow$ global smoothness is Theorem A and Section 10. Definition 7.1 for arbitrary data is Theorem C and is not proved.

1 Scope

This note treats the unmodified Cauchy problem for 3D incompressible Navier–Stokes, written as (1)–(2) below, on the maximal classical existence interval $[0, T^*)$. The only geometric input is a uniform spatial $\frac{1}{2}$-Hölder bound on the vorticity *direction* on the set where the vorticity *magnitude* is at least a positive threshold. From that input one concludes $T^* = \infty$.

The following are **not** proved: that $C(t)$ remains finite up to T^* for every smooth divergence-free u_0; that $T^* < \infty$ occurs for some such u_0; that a named tether or metriplectic correction is present in the equations. The PDE used is (1).

Section 9 constructs candidate constants from the PDE and tests whether the energy identity forces them to stay finite. It does not.

2 The equations

Let $\nu > 0$. On $\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$ (the whole-space case is identical up to replacing the periodized Newtonian potential by $1/(4\pi|x|)$) one considers

$$\partial_t u + (u \cdot \nabla)u = \nu\Delta u - \nabla p, \tag{1}$$

$$\nabla \cdot u = 0, \tag{2}$$

with $u(\,\cdot\,, 0) = u_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$, $\nabla \cdot u_0 = 0$. Local well-posedness in H^s, $s > 5/2$, yields a unique classical solution on a maximal interval $[0, T^*)$, $T^* \in (0, \infty]$, with the standard blow-up alternative: if $T^* < \infty$ then

$$\limsup_{t \uparrow T^*} \|\nabla u(t)\|_{L^2(\mathbb{T}^3)} = \infty. \tag{3}$$

Equivalently, $\limsup_{t\uparrow T^*} \|\omega(t)\|_{L^2} = \infty$, since $\|\omega\|_{L^2} = \|\nabla u\|_{L^2}$ under (2). This is the energy-class continuation criterion (Kato); it is the only continuation tool used below. (Beale–Kato–Majda in $L^1_t L^\infty_x$ is stronger than needed once enstrophy is bounded.)

Curl of (1), using (2), gives the vorticity equation

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu\Delta\omega, \qquad \omega = \nabla \times u. \tag{4}$$

Write $r := |\omega|$ and, on the open set $\{r > 0\}$,

$$\xi := \frac{\omega}{r}. \tag{5}$$

The strain is the symmetric gradient

$$S := \frac{1}{2}\big(\nabla u + (\nabla u)^\top\big). \tag{6}$$

On $\{r > 0\}$ one has the stretching identity

$$(\omega \cdot \nabla)u \cdot \omega = r^2\,(S\xi \cdot \xi). \tag{7}$$

The left-hand side of (7) extends continuously by 0 at $r = 0$. The unit field ξ is *not* extended; every estimate below that mentions $\xi(x)$ is restricted to $r(x) > 0$.

3 The two constants

The geometric criterion uses exactly two solution-class constants, both allowed to depend on the data (u_0, ν) and on nothing else from the later evolution:

$$\Omega = \Omega(u_0, \nu) > 0, \tag{8}$$
$$C = C(u_0, \nu) < \infty. \tag{9}$$

Here Ω is the vorticity-intensity threshold: ξ is constrained only on $\{r \geq \Omega\}$. The complementary set $\{r < \Omega\}$ is estimated without forming ξ. The constant C is the uniform spatial $\frac{1}{2}$-Hölder modulus of ξ on that high set. From (8)–(9) the enstrophy argument produces one derived majorant coefficient, depending on the two constants and on viscosity,

$$K(C, \Omega, \nu) = \frac{C_*}{\nu}\,(C + \Omega)^2, \tag{10}$$

where $C_* < \infty$ depends only on the Sobolev, Hardy–Littlewood–Sobolev, and Calderón–Zygmund constants of $\mathbb{T}^3$ (or $\mathbb{R}^3$). The pair (Ω, C) is the input; K is not an independent hypothesis.

These two constants are not constructed from (1) in this note. Theorem A takes (8)–(9) as given.

4 The criterion

Theorem A (Beirão da Veiga–Berselli, recast)**.** *Let u be the classical solution of* (1)–(2) *on* $[0, T^*)$, *and let $\Omega = \Omega(u_0, \nu)$ and $C = C(u_0, \nu)$ be the two constants* (8)–(9). *Suppose that for all $t \in [0, T^*)$ and all $x, y \in \mathbb{T}^3$ with $r(x, t) \geq \Omega$ and $r(y, t) \geq \Omega$,*

$$|\xi(x, t) - \xi(y, t)| \leq C\,|x - y|^{1/2}. \tag{11}$$

Then $T^* = \infty$, and $u \in C^\infty(\mathbb{T}^3 \times [0, \infty))$. The continuation bound depends on the two constants only through $K(C, \Omega, \nu)$ in (10)*:*

$$\sup_{t<T^*} \|\omega(t)\|_{L^2}^2 \leq \|\omega(0)\|_{L^2}^2 \exp\Big(\frac{K(C, \Omega, \nu)}{2\nu}\,\|u_0\|_{L^2}^2\Big). \tag{12}$$

The Hölder condition may be localized to $|x - y| < \delta$ for an arbitrary $\delta > 0$; the far field is controlled by the integrable periodized kernel. Only pairs in which *both* magnitudes meet the threshold Ω are constrained. Pairs with $r(x) < \Omega$ or $r(y) < \Omega$ are estimated crudely, without using ξ.

Remark 1. On $\{r(x), r(y) > 0\}$ one has $|\xi(x) - \xi(y)| = 2|\sin(\theta/2)|$ and $\sin\theta \leq |\xi(x) - \xi(y)|$, where $\theta = \angle(\omega(x), \omega(y))$. Thus (11) implies $\sin\theta(x, y, t) \leq C|x - y|^{1/2}$ on the high-high set, which is Assumption A of Beirão da Veiga–Berselli [1] at exponent $\alpha = 1/2$ with g a constant. Lipschitz direction (Constantin–Fefferman [3]) is the case $\alpha = 1$.

5 Enstrophy and stretching

Take the L^2 inner product of (4) with ω. After an integration by parts (justified on every compact subinterval of $[0, T^*)$ by classical smoothness),

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 + \nu\|\nabla\omega\|_{L^2}^2 = \int_{\mathbb{T}^3} \mathcal{K}(x,t)\,\mathrm{d}x, \tag{13}$$

where the stretching density is

$$\mathcal{K}(x,t) := \big((\omega\cdot\nabla)u\cdot\omega\big)(x,t). \tag{14}$$

On $\{r = 0\}$ one has $\mathcal{K} = 0$. On $\{r > 0\}$ one has $\mathcal{K} = r^2(S\xi\cdot\xi)$.

Biot–Savart on $\mathbb{T}^3$ reconstructs u from ω by the periodized Newtonian potential G. Differentiating under the principal value gives S as a Calderón–Zygmund operator of ω. The Constantin–Fefferman representation [3] then yields the pointwise bound, for a dimensional constant c_0 independent of u,

$$|\mathcal{K}(x)| \le c_0 \int_{\mathbb{T}^3} \frac{r(x)^2\, r(y)\, \sin\theta(x,y)}{|x-y|^3}\,\mathrm{d}y, \tag{15}$$

where the integrand is read as 0 if $r(x) = 0$ or $r(y) = 0$, so that ξ is never invoked on the degeneracy set. (On the torus, $|x-y|$ denotes chordal/periodized distance; the local singularity is the same as in $\mathbb{R}^3$.)

6 High-high versus complementary pairs

Fix $\Omega > 0$ as in Theorem A. Split the integral in (15) according to whether $r(y) \ge \Omega$ or not. Write $\mathcal{K} = \mathcal{K}_{\mathrm{hh}} + \mathcal{K}_{\mathrm{c}}$.

High-high. If $r(x) \ge \Omega$ and $r(y) \ge \Omega$, (11) and $\sin\theta \le |\xi(x) - \xi(y)|$ give $\sin\theta/|x-y|^3 \le C|x-y|^{-5/2}$. Hence

$$|\mathcal{K}_{\mathrm{hh}}(x)| \le c_0 C\, r(x)^2\, I(x), \qquad I(x) := \int_{\mathbb{T}^3} \frac{r(y)}{|x-y|^{5/2}}\,\mathrm{d}y. \tag{16}$$

The field I is a Riesz potential of order $\beta = 1/2$ applied to r. Hardy–Littlewood–Sobolev on $\mathbb{T}^3$ (or $\mathbb{R}^3$) yields

$$\|I\|_{L^3} \le C_{\mathrm{HLS}}\|\omega\|_{L^2}, \tag{17}$$

because $1/3 = 1/2 - (1/2)/3$. If $r(x) < \Omega$, the factor $r(x)^2$ already places $\mathcal{K}_{\mathrm{hh}}(x)$ in the complementary class treated next; alternatively one may restrict (16) to $\{r(x) \ge \Omega\}$ and estimate the remainder with $\sin\theta \le 1$ and $r(x) < \Omega$.

Complementary pairs. If $r(y) < \Omega$, use $\sin\theta \le 1$ and do *not* form $\xi(y)$. After the usual cancellation in the CZ kernel of S (mean-zero homogeneous kernel of degree -3), the complementary stretching is controlled by the Hardy–Littlewood maximal function of the truncated vorticity:

$$|\mathcal{K}_{\mathrm{c}}(x)| \le c_1 r(x)^2\, M\big(r\,1_{\{r<\Omega\}}\big)(x). \tag{18}$$

In particular, using $r1_{\{r<\Omega\}} \le \Omega$ and the boundedness of $M : L^2 \to L^2$,

$$\int_{\mathbb{T}^3} |\mathcal{K}_{\mathrm{c}}|\,\mathrm{d}x \le c_2\Omega\,\|\omega\|_{L^2}\|\omega\|_{L^6}. \tag{19}$$

(The same bound follows from $|S(1_{\{r<\Omega\}}\omega)| \lesssim M(r1_{\{r<\Omega\}})$. No division by r occurs.)

7 Closing the enstrophy inequality

Combining (13), (16), (17) and (19),

$$\int |\mathcal{K}_{\mathrm{hh}}| \,\mathrm{d}x \le c_0 C \int r^2 I \,\mathrm{d}x \le c_0 C \|\omega\|_{L^3}^2 \|I\|_{L^3} \le c_3 C \|\omega\|_{L^2} \|\omega\|_{L^6} \|\omega\|_{L^2},$$

where the interpolation $\|\omega\|_{L^3}^2 \le \|\omega\|_{L^2}\|\omega\|_{L^6}$ was used. Thus

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 + \nu\|\nabla\omega\|_{L^2}^2 \le c_4(C+\Omega)\,\|\omega\|_{L^2}^2\|\omega\|_{L^6}. \tag{20}$$

Sobolev on $\mathbb{T}^3$ gives $\|\omega\|_{L^6} \le C_{\mathrm{S}}\|\nabla\omega\|_{L^2}$. Young's inequality, for every $\varepsilon > 0$,

$$\|\omega\|_{L^2}^2\|\omega\|_{L^6} \le \frac{\varepsilon}{c_4(C+\Omega)}\|\omega\|_{L^6}^2 + \frac{c_4(C+\Omega)}{4\varepsilon}\|\omega\|_{L^2}^4,$$

so choosing ε small enough to absorb the first term into $\nu\|\nabla\omega\|_{L^2}^2$ produces

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 + \nu\|\nabla\omega\|_{L^2}^2 \le K(C,\Omega,\nu)\,\|\omega\|_{L^2}^4, \tag{21}$$

with $K(C,\Omega,\nu)$ the derived coefficient (10). The only data-dependent constants entering K are the two constants Ω and C.

8 Continuation

Energy equality for (1) on $[0,T^*)$ yields

$$\frac{1}{2}\|u(t)\|_{L^2}^2 + \nu\int_0^t \|\nabla u(s)\|_{L^2}^2\,\mathrm{d}s = \frac{1}{2}\|u_0\|_{L^2}^2, \tag{22}$$

hence $\int_0^t \|\omega(s)\|_{L^2}^2\,\mathrm{d}s \le (2\nu)^{-1}\|u_0\|_{L^2}^2$ for every $t < T^*$. Drop the dissipation in (21) and apply Gronwall:

$$\|\omega(t)\|_{L^2}^2 \le \|\omega(0)\|_{L^2}^2 \exp\Big(K\int_0^t \|\omega(s)\|_{L^2}^2\,\mathrm{d}s\Big) \le \|\omega(0)\|_{L^2}^2 \exp\big(K(2\nu)^{-1}\|u_0\|_{L^2}^2\big). \tag{23}$$

The right-hand side is independent of $t < T^*$. Therefore $\sup_{t<T^*}\|\nabla u(t)\|_{L^2} < \infty$. The blow-up alternative (3) forces $T^* = \infty$.

Once the solution exists globally in H^1 with uniformly bounded enstrophy, standard parabolic bootstrapping for (1) (Kato, or L^p-L^q estimates for the Stokes semigroup) upgrades u to $C^\infty(\mathbb{T}^3 \times [0,\infty))$.

This is Theorem A.

9 Construction from the PDE, and the test

The two constants are now built from (1)–(2) and from u_0, then tested against the energy identity. No term is added to the PDE. Degeneracy of ξ on $\{r = 0\}$ is respected.

9.1 Irrotational data

If $\omega_0 \equiv 0$, then u_0 is a harmonic divergence-free field on $\mathbb{T}^3$, hence constant. The solution of (1)–(2) is $u(\,\cdot\,,t) = u_0$, which is smooth for all time. Take $\Omega = 1$ and $C = 0$; the high set is empty. This case is settled and is excluded below.

9.2 The threshold Ω

Fix the intensity threshold independently of the later evolution,

$$\Omega(u_0, \nu) := 1. \tag{24}$$

Any other fixed positive number may replace 1; Beirão da Veiga–Berselli allow an arbitrary $k > 0$. The choice must *not* track $\|\omega(t)\|_\infty$. In particular one cannot take $\Omega(t) > \|\omega(t)\|_\infty$, which would empty the high set and make (11) vacuous by circular use of a bound on ω. The value (24) depends on (u_0, ν) only in that $\nu > 0$ and u_0 is given; it is a universal positive threshold.

9.3 The modulus $C(t)$ along the classical solution

Let $E(t) := \{x \in \mathbb{T}^3 : r(x,t) \geq \Omega\}$. On each compact subinterval of $[0, T^*)$ the solution is smooth, so $E(t)$ is closed and $\xi(t)$ is smooth on a neighbourhood of $E(t)$ whenever $E(t)$ is nonempty (because $r \geq 1$ there). Define the running $\frac{1}{2}$-Hölder modulus on the high set by

$$C(t) := \begin{cases} 0 & \text{if } E(t) = \emptyset, \\ \sup\left\{ \frac{|\xi(x,t) - \xi(y,t)|}{|x-y|^{1/2}} : x, y \in E(t),\ x \neq y \right\} & \text{if } E(t) \neq \emptyset. \end{cases} \tag{25}$$

Local smoothness gives $C(t) < \infty$ for every $t < T^*$. The constant required by Theorem A is the *uniform* value

$$C(u_0, \nu) := \sup_{t \in [0, T^*)} C(t) \in [0, \infty]. \tag{26}$$

This is constructed from the PDE in the only non-circular sense available: it is the actual modulus of the classical solution on its maximal interval. Theorem A applies if and only if this supremum is finite.

9.4 Direction equation on $\{r > 0\}$

On $\{r > 0\}$, $|\xi| = 1$ and $(\omega \cdot \nabla)u = S\omega$. The chain rule in (4) yields

$$D_t \xi = P_\xi(S\xi) + \nu\, r^{-1} P_\xi(\Delta\omega), \tag{27}$$

where $D_t = \partial_t + u \cdot \nabla$ and $P_\xi v := v - (\xi \cdot v)\xi$. Expand $\Delta\omega = \Delta(r\xi)$:

$$\Delta\omega = (\Delta r)\xi + 2\nabla r \cdot \nabla\xi + r\Delta\xi.$$

Then $P_\xi(\Delta\omega) = 2P_\xi(\nabla r \cdot \nabla\xi) + rP_\xi(\Delta\xi)$. The constraint $|\xi| = 1$ gives $\xi \cdot \nabla\xi = 0$ and $\xi \cdot \Delta\xi = -|\nabla\xi|^2$, hence $P_\xi(\Delta\xi) = \Delta\xi + |\nabla\xi|^2\xi$. Substituting produces the correct viscous equation

$$D_t\xi - \nu\Delta\xi = P_\xi(S\xi) + 2\nu r^{-1} P_\xi(\nabla r \cdot \nabla\xi) + \nu|\nabla\xi|^2\xi. \tag{28}$$

The drift is projected: it is $P_\xi(\nabla r \cdot \nabla\xi)$, not $\nabla r \cdot \nabla\xi$. The quadratic term $\nu|\nabla\xi|^2\xi$ is the second fundamental form of S^2 (harmonic-map tension) and must be kept. Componentwise $\Delta\xi + 2\nabla \log r \cdot \nabla\xi = r^{-2}\mathrm{div}(r^2\nabla\xi)$, but the equation uses the projection of that operator, not the unprojected drift.

On $E(t) = \{r \geq \Omega\}$ one has $|\nabla\xi| \leq \Omega^{-1}|\nabla\omega|$ and $|\nabla r| \leq |\nabla\omega|$ (Pythagoras $|\nabla\omega|^2 = |\nabla r|^2 + r^2|\nabla\xi|^2$). In particular $|\nabla r| \leq r$ is false, and $|\nabla \log r| \leq \Omega^{-1}|\nabla\omega|$ is not a bound independent of the solution: distance to $\partial E(t)$ does not control $|\nabla r|$ inside $E(t)$. The drift is therefore not $O(1/\Omega)$ in any uniform sense. No division by r occurs on $\{r = 0\}$.

9.5 Morrey reduction

Morrey's embedding on $\mathbb{T}^3$ gives $W^{1,6} \hookrightarrow C^{0,1/2}$. Let $\chi \in C^\infty(\mathbb{R})$ satisfy $\chi(s) = 0$ for $s \le 1$ and $\chi(s) = 1$ for $s \ge 2$, and set $v(x,t) := \chi\big(r(x,t)/\Omega\big)\,\xi(x,t)$ on $\{r > 0\}$, with $v = 0$ on $\{r = 0\}$. Then v is a globally defined smooth field on $[0, T^*)$, $v = \xi$ on $\{r \ge 2\Omega\}$, and

$$|\nabla v| \le C_\chi\,\Omega^{-1}|\nabla\omega|. \tag{29}$$

Hence a finite bound

$$\sup_{t<T^*} \|\nabla\omega(t)\|_{L^6(\mathbb{T}^3)} < \infty \tag{30}$$

would imply $\sup_{t<T^*} C(t) < \infty$ after replacing Ω by 2Ω in Theorem A. Lipschitz control of ξ would require $\nabla\omega \in L^\infty_t L^\infty_x$, which is Beale–Kato–Majda and is strictly stronger than (30).

9.6 What the energy identity supplies

Energy equality (22) yields only

$$u \in L^\infty(0,T^*;L^2) \cap L^2(0,T^*;H^1), \qquad \omega \in L^2(0,T^*;L^2).$$

It does *not* yield $\nabla\omega \in L^\infty_t L^6$, nor even $\nabla\omega \in L^2_t L^2$. The latter would be an enstrophy bound, which is what Theorem A is designed to produce after C is known to be finite. Using (30) as a hypothesis to bound C is therefore circular at the energy level.

A differential inequality for $C(t)$ from (27) contains the stretching $P_\xi(S\xi)$ and the viscous term $\nu\Omega^{-1}P_\xi(\Delta\omega)$ on $E(t)$. The strain S is a Calderón–Zygmund image of ω. Neither term is controlled in $L^\infty_t L^6$ by $\|\omega\|_{L^2_t L^2_x}$ alone. The same obstruction persists if $\Omega = 1$ is replaced by any other fixed positive threshold depending only on $\|u_0\|_{H^s}$ and ν.

9.7 Result of the test

Theorem B (construction from the PDE does not close)**.** *Let $u_0 \in C^\infty(\mathbb{T}^3;\mathbb{R}^3)$ be divergence-free and $\nu > 0$, and let u be the classical solution of* (1)–(2) *on* $[0,T^*)$*. Define Ω by* (24) *and $C(u_0,\nu)$ by* (26)*. Then:*

1. *If $\omega_0 \equiv 0$, then $T^* = \infty$ and u is globally smooth.*
2. *If $\omega_0 \not\equiv 0$, then $C(t) < \infty$ for every $t < T^*$, but the energy identity for* (1) *does not imply $C(u_0,\nu) < \infty$. In particular it does not imply* (30)*.*
3. *Consequently Theorem A cannot be applied from the energy class alone. The test does not prove $T^* = \infty$, and it does not produce a finite-time singularity.*

Local smoothness on compact subintervals of $[0,T^*)$ always gives a finite modulus on each such interval. Uniformity up to T^* is the whole problem. Values $\beta < 1/2$ are not known to suffice [2].

No term was added to (1). Degeneracy of ξ on $\{r = 0\}$ was respected throughout. The Clay problem, as stated by Fefferman [4], remains open.

10 The geometric implication: Definition 7.1 $\Rightarrow$ Clay

This section is the Constantin–Fefferman / Beirão da Veiga–Berselli implication, written so that it closes. If Definition 7.1 holds for every smooth divergence-free u_0, the answer to the Clay problem is yes. Definition 7.1 is not proved here; it is Theorem C.

Definition 7.1 (geometric criterion). There exist $\Omega = \Omega(u_0, \nu) > 0$ and $C = C(u_0, \nu) < \infty$ such that for all $t \in [0, T^*)$ and all x, y with $r(x,t) \geq \Omega$ and $r(y,t) \geq \Omega$,

$$|\xi(x,t) - \xi(y,t)| \leq C\,|x-y|^{1/2}.$$

In particular $\sin\phi \leq C|y|^{1/2}$ on that high-high set, where $\phi = \angle(\omega(x), \omega(x+y))$. The constants C, Ω do not depend on T^*.

Definition 7.2 (continuation). If $T^* < \infty$ then $\limsup_{t\uparrow T^*} \|\nabla u(t)\|_{L^2} = \infty$ (Kato H^1). Equivalently $\limsup_{t\uparrow T^*} \|\omega(t)\|_{L^2} = \infty$. Beale–Kato–Majda ($\int_0^{T^*} \|\omega\|_{L^\infty} = \infty$) is a stronger necessary condition for blow-up; it is not the criterion fed by an L^2 enstrophy bound. Once $T^* = \infty$, Beale–Kato–Majda holds automatically on every finite interval.

Theorem (geometric regularity). Let $u_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$, $\nabla \cdot u_0 = 0$, $\nu > 0$, and let u be the unique classical solution of (1)–(2) on $[0, T^*)$. Assume Definition 7.1. Then $T^* = \infty$ and $u \in C^\infty([0,\infty) \times \mathbb{T}^3)$.

Proof. *1. Energy.*

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|u\|_{L^2}^2 + \nu\|\nabla u\|_{L^2}^2 = 0,$$

so $\|u(t)\|_{L^2} \leq \|u_0\|_{L^2}$ for $t < T^*$. On $\mathbb{T}^3$, $\|\omega\|_{L^2} \simeq \|\nabla u\|_{L^2}$.

2. Enstrophy identity. Curl of (1) and the L^2 pairing give

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 = \int_{\mathbb{T}^3} (\xi \cdot S\xi)\, r^2 \,\mathrm{d}x - \nu\|\nabla\omega\|_{L^2}^2. \tag{31}$$

The stretching density is $\mathcal{K} = r^2(\xi \cdot S\xi)$, not $\xi \cdot S\xi$. On $\{r = 0\}$ one has $\mathcal{K} = 0$. Pythagoras: $|\nabla\omega|^2 = |\nabla r|^2 + r^2|\nabla\xi|^2$.

3. Strain as a geometric kernel. Biot–Savart gives the Constantin–Fefferman representation

$$(\xi \cdot S\xi)(x) = \frac{3}{4\pi}\,\mathrm{P.V.}\int_{\mathbb{T}^3} D\big(\hat{y}, \xi(x), \xi(x+y)\big)\frac{r(x+y)}{|y|^3}\,\mathrm{d}y,$$

with $D(\hat{y}, \xi, \xi') = (\hat{y}\cdot\xi)\det(\hat{y}, \xi', \xi)$ and $|D| \leq |\sin\phi(x, x+y)| \leq |\xi(x) - \xi(x+y)|$. If $\xi(x) = \xi(x+y)$ then $D = 0$: aligned vorticity does not stretch. That is the geometry.

4. Stretching under Definition 7.1. Split $\int(\xi \cdot S\xi)r^2$ into three pieces.

Low vorticity ($r(x) < \Omega$). Then $r^2 < \Omega^2$, so

$$\Big|\int_{\{r<\Omega\}} (\omega\cdot\nabla)u\cdot\omega\,\mathrm{d}x\Big| \leq \Omega^2\int|\nabla u| \leq C_{\mathbb{T}}\Omega^2\|\omega\|_{L^2}.$$

No $\|\omega\|_{L^\infty}$. Degeneracy: ξ is not used on this set.

Far interactions ($|y| \geq 1$). The kernel is smooth. Integrating by parts against $\omega = \nabla \times u$, the far strain is a bounded operator on u. Energy from Step 1 gives

$$|I_{\text{far}}| \leq K_{\text{far}}\|u_0\|_{L^2}\|\omega\|_{L^2}^2.$$

Near high-high ($|y| < 1$ and $r(x), r(x+y) \geq \Omega$). Definition 7.1 gives $|D| \leq C|y|^{1/2}$, so the kernel is $|y|^{-5/2}$. In three dimensions

$$\int_{|y|<1} |y|^{-5/2}\,\mathrm{d}y = \int_0^1 \rho^{-1/2}\,\mathrm{d}\rho < \infty :$$

the singularity is integrable. Hölder $1/2$ is critical for this integrability (exponent $< 1/2$ makes the radial integral diverge). The contribution is $\int r(x)^2 I(x)\,\mathrm{d}x$ with $I = r * |\cdot|^{-5/2}$. Hardy–Littlewood–Sobolev at order $1/2$ yields $\|I\|_{L^3} \le C_{\mathrm{HLS}}\|\omega\|_{L^2}$. Interpolation $\|\omega\|_{L^3}^2 \le \|\omega\|_{L^2}\|\omega\|_{L^6}$ then gives

$$\int r^2 I \le C_{\mathrm{HLS}} C\,\|\omega\|_{L^2}^2 \|\omega\|_{L^6}. \tag{32}$$

Sobolev $\|\omega\|_{L^6} \le C_{\mathrm{S}}\|\nabla\omega\|_{L^2}$ and Young with $\varepsilon = \nu/2$ produce

$$c\,\|\omega\|_{L^2}^2\|\omega\|_{L^6} \le \frac{\nu}{2}\|\nabla\omega\|_{L^2}^2 + C_\nu(C)\,\|\omega\|_{L^2}^4.$$

The exponent 4 is the one that closes. A bound $\|\omega\|_{L^2}^{5/2}\|\nabla\omega\|_{L^2}^{1/2}$ would Young-close to $\|\omega\|_{L^2}^{10/3}$, and $y' \le Ky^{5/3}$ for $y = \|\omega\|_{L^2}^2$ can blow up in finite time; that interpolation does not close. The chain (32) gives $y' \le Ky^2$, which Gronwall controls from energy.

5. Enstrophy inequality. Summing the three pieces,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 + \nu\|\nabla\omega\|_{L^2}^2 \le K(C,\Omega,\nu)\,\|\omega\|_{L^2}^4 + C'\Omega^2\|\omega\|_{L^2}, \tag{33}$$

with $K = C_*(C+\Omega)^2/\nu$ as in (10). The right-hand side is not a constant independent of ω. Energy supplies $\int_0^t \|\omega\|_{L^2}^2\,\mathrm{d}s \le (2\nu)^{-1}\|u_0\|_{L^2}^2$. Gronwall yields, for every $t < T^*$,

$$\|\omega(t)\|_{L^2}^2 \le \big(\|\omega_0\|_{L^2}^2 + C''\big)\exp\Big(\frac{K}{2\nu}\|u_0\|_{L^2}^2\Big).$$

The exponential does not depend on t or on T^*, because C and Ω in Definition 7.1 do not. Hence $\omega \in L^\infty(0,T^*;L^2) \cap L^2(0,T^*;H^1)$, i.e. $u \in L^\infty(0,T^*;H^1(\mathbb{T}^3))$.

6. Serrin class and continuation. On $\mathbb{T}^3$, $H^1 \hookrightarrow L^6$, so $u \in L^\infty(0,T^*;L^6)$. Prodi–Serrin–Ladyzhenskaya: a Leray–Hopf solution in $L_t^s L_x^q$ with $3/q + 2/s \le 1$ and $q > 3$ is smooth. Here $q = 6$, $s = \infty$, and $3/6 + 2/\infty = 1/2 < 1$, which is subcritical. Thus if $T^* < \infty$, u is smooth on $[0,T^*]$. Kato local existence: a smooth H^s solution ($s > 5/2$) with finite H^s norm at time T^* extends to $T^* + \varepsilon$. That contradicts maximality. Therefore $T^* = \infty$.

This is Kato–Fujita continuation in H^1, not an application of Beale–Kato–Majda to an L^2 bound. An L^2 bound on ω does not control $\int\|\omega\|_{L^\infty}$. It does control $\|u\|_{H^1}$, which is the continuation class used here. Beale–Kato–Majda is then automatic on every finite interval.

7. Global smoothness. Parabolic regularity for (1) on $\mathbb{T}^3$: a solution that is smooth on every compact time interval, with $T^* = \infty$, is C^∞ for all $t \ge 0$.

Thus Definition 7.1 $\Rightarrow$ Definition 7.2 cannot fire $\Rightarrow T^* = \infty \Rightarrow$ global smoothness. Alignment depletes stretching ($D = 0$ when $\xi(x) = \xi(x+y)$). Hölder $1/2$ makes the leftover kernel integrable at the origin and matches HLS with $\omega \in L^2$. The resulting bound on enstrophy is independent of T^* because C, Ω are. Bounded enstrophy is $L_t^\infty H^1$, a subcritical Serrin class. Smooth solutions in that class cannot stop at a finite T^*.

That implication is a complete geometric solution of Clay if and only if Definition 7.1 holds for every smooth divergence-free u_0. That last statement is Theorem C and is not proved here.

11 What is left

Everything in Theorems A–B is done. One statement remains. If it is proved, Theorem A gives $T^* = \infty$ for every smooth divergence-free u_0, which is the Clay problem [4]. If it fails for some u_0, this route does not decide blow-up; a different criterion would be needed.

11.1 The remaining statement

Theorem C (remaining; not proved)**.** *Let* $u_0 \in C^\infty(\mathbb{T}^3;\mathbb{R}^3)$ *be divergence-free,* $\nu > 0$*, and* u *the classical solution of* (1)–(2) *on* $[0, T^*)$*. Set* $\Omega = 1$ *as in* (24)*. Then*

$$C(u_0,\nu) := \sup_{t\in[0,T^*)} C(t) < \infty, \tag{34}$$

with $C(t)$ *as in* (25)*. The finite value may depend on* (u_0, ν) *only.*

That is the entire remainder for this route. Equivalently, writing $E(t) = \{r(\,\cdot\,, t) \geq 1\}$,

$$\sup_{t<T^*} \sup_{\substack{x,y\in E(t)\\ x\neq y}} \frac{|\xi(x,t) - \xi(y,t)|}{|x-y|^{1/2}} < \infty. \tag{35}$$

No other estimate is missing from Theorem A: local existence, the stretching representation, the high-high / complementary split, Hardy–Littlewood–Sobolev at order 1/2, absorption into $\nu\|\nabla\omega\|^2_{L^2}$, Gronwall against the energy integral, and the H^1 blow-up alternative are already closed.

11.2 A legitimate reduction, and two that are not

On $E(t)$ one has $r \geq 1$, hence

$$|\xi(x) - \xi(y)| \leq 2|\omega(x) - \omega(y)|. \tag{36}$$

Thus (34) follows from uniform $\frac{1}{2}$-Hölder continuity of ω itself on $E(t)$. That is still a statement only about intense vorticity, not about $\|\omega\|_\infty$.

The following are *not* the remaining problem, because each is already known to be equivalent to global regularity by continuation, and using any of them to bound C is circular:

- $\sup_{t<T^*}\|\omega(t)\|_{L^2} < \infty$ (Kato H^1),
- $\int_0^{T^*} \|\omega(t)\|_{L^\infty}\,\mathrm{d}t < \infty$ (Beale–Kato–Majda),
- $\nabla\omega \in L^\infty(0,T^*; L^6)$ (Morrey, (30)),
- $u \in L^p_t L^q_x$ with $2/p + 3/q = 1$, $q > 3$ (Serrin).

The point of (34) is that it is geometrically weaker than a bound on $|\omega|$: a vortex that intensifies while remaining spatially aligned can have large r and still small $|\xi(x) - \xi(y)|$. Theorem A is designed to turn that alignment into an enstrophy bound. Beirão da Veiga [2] argues that $\beta = 1/2$ is critical for this method; the remaining statement is not a minor calculation.

11.3 What the direction equation must yield

The only unused dynamical information is (27) on $E(t)$. For $x, y \in E(t)$,

$$\begin{aligned} D^x_t\xi(x) - D^y_t\xi(y) = {} & P_{\xi(x)}\big(S(x)\xi(x)\big) - P_{\xi(y)}\big(S(y)\xi(y)\big) \\ & + \nu\Big(r(x)^{-1}P_{\xi(x)}\Delta\omega(x) - r(y)^{-1}P_{\xi(y)}\Delta\omega(y)\Big), \end{aligned} \tag{37}$$

where $D^x_t = \partial_t + u(x)\cdot\nabla_x$. The first line is the misalignment of strain; the second is the viscous increment of direction. To prove (34) it is necessary and sufficient to show that this difference, divided by $|x-y|^{1/2}$, remains bounded on $E(t)\times E(t)$ up to T^*, with a bound depending only on (u_0,ν). Degeneracy is already removed: $r(x), r(y) \geq 1$.

That difference estimate is the next (and last) step on this route.

11.4 Beginning of the estimate for Definition 7.1

Fix $\Omega = 1$. For $x, y \in E(t)$ the elementary identity

$$|\xi(x) - \xi(y)|^2 = 2\big(1 - \xi(x) \cdot \xi(y)\big) = 4 \sin^2(\phi/2) \tag{38}$$

identifies the Hölder modulus of ξ with the CF angle in the kernel D. Thus Definition 7.1 is exactly the statement that the geometric factor $|D|$ is $O(|y|^{1/2})$ uniformly on high-high pairs, with a constant independent of $t < T^*$.

On $E(t)$ one also has (36). Along any segment that remains in $\{r \geq 1/2\}$,

$$\xi(x) - \xi(y) = \int_0^1 (x - y) \cdot \nabla \xi\big(y + s(x - y)\big)\, \mathrm{d}s, \tag{39}$$

and $|\nabla \xi| \leq 2|\nabla \omega|$ there. Bounding the right-hand side by $|x - y|\, \|\nabla \omega\|_{L^\infty}$ recovers a Lipschitz estimate, which is Beale–Kato–Majda, hence circular. Definition 7.1 asks for one derivative less: a factor $|x - y|^{1/2}$ rather than $|x - y|$. The unused dissipation in Pythagoras,

$$|\nabla \omega|^2 = |\nabla r|^2 + r^2 |\nabla \xi|^2,$$

puts $|\nabla \xi|$ in L^2 only after enstrophy is bounded, which is again circular. The estimate that would close Definition 7.1 without a derivative too many is a Campanato bound on $E(t)$,

$$\sup_{x \in E(t)} \sup_{\rho \in (0,1)} \rho^{-4} \int_{B_\rho(x) \cap E(t)} |\xi(z) - \xi_{x,\rho}|^2\, \mathrm{d}z \leq C(u_0, \nu)^2, \tag{40}$$

equivalent to (34) on the high set. The integrand is controlled by $\sin^2(\phi/2)$ via (38), hence by the same D that depletes stretching. Whether (40) follows from (27) and energy, with no bound on $\|\omega\|_{L^\infty}$, is the whole of Definition 7.1.

No other analytic ingredient is missing. Once (40) holds, Section 10 is unconditional and $T^* = \infty$.

11.5 Corrected De Giorgi–Nash–Moser attempt

A proposed parabolic De Giorgi–Nash–Moser / Campanato argument for Definition 7.1 is rewritten here with the necessary corrections. It does not prove Definition 7.1. The surviving facts are recorded, then the gaps.

Forcing from Calderón–Zygmund $2 \to 2$, without $\|\omega\|_{L^\infty}$. The pointwise bound $|S| \leq C|\omega|$ requires $\|\omega\|_{L^\infty}$ and is not used. Energy for (1) gives $\omega \in L^2(0, T^*; L^2)$ (equivalently $\nabla u \in L^2_t L^2_x$). The strain is a Calderón–Zygmund operator of ω, hence $S \in L^2(0, T^*; L^2)$ by the $L^2 \to L^2$ bound, globally, with no pointwise kernel estimate. Then $|P_\xi(S\xi)| \leq |S|$, so

$$P_\xi(S\xi) \in L^2(0, T^*; L^2(\mathbb{T}^3)). \tag{41}$$

That term of (28) is under control.

Drift is not absorbed by Leray–Hopf. On a space-time cylinder $Q \subset \{r \geq \Omega\}$,

$$|2\nu r^{-1} P_\xi(\nabla r \cdot \nabla \xi)| \leq 2\nu \Omega^{-1} |\nabla r|\, |\nabla \xi| \leq 2\nu \Omega^{-2} |\nabla \omega|^2,$$

using $|\nabla r| \leq |\nabla \omega|$ and $|\nabla \xi| \leq \Omega^{-1} |\nabla \omega|$. (The bound $|\nabla r| \leq |\omega|$ is false: Pythagoras gives $|\nabla r| \leq |\nabla \omega|$, not $|\nabla r| \leq r$.) Thus the drift lies in $L^2_t L^2_x(Q)$ only if $\nabla \omega \in L^4(Q)$. Leray–Hopf energy does not provide $\nabla \omega \in L^2$, let alone L^4. Ladyzhenskaya interpolation of $|\omega|^2 |\nabla \omega|^2$ against $|\nabla \omega|^4$ and $|\omega|^4$ likewise requires norms of ω that energy does not give. There is no energy inequality for ω in the Leray–Hopf class; that would be an enstrophy bound, which is the output of Definition 7.1 rather than an input. The drift term in (28) is therefore not known to lie in $L^2_t L^2_x$ on cylinders in $\{r \geq \Omega\}$.

Quadratic tension. The term $\nu|\nabla\xi|^2\xi$ satisfies $|\nabla\xi|^2 \le \Omega^{-2}|\nabla\omega|^2$ on $E(t)$. Membership in $L^2_tL^2_x$ would require $\nabla\xi \in L^4$, i.e. again $\nabla\omega \in L^4$. Intrinsically, this term is normal to $T_\xi S^2$. The tangential equation is

$$P_\xi(D_t\xi - \nu\Delta\xi) = P_\xi(S\xi) + 2\nu r^{-1}P_\xi(\nabla r \cdot \nabla\xi), \tag{42}$$

which is the harmonic-map heat flow into S^2 with an L^2 stretching force and an uncontrolled drift. Finite-time singularities of the harmonic-map heat flow from three-dimensional domains into S^2 are known; quadratic growth in $\nabla\xi$ is not a perturbation of a scalar De Giorgi–Nash–Moser equation.

What De Giorgi–Nash–Moser would require, and does not have. Scalar parabolic De Giorgi–Nash–Moser needs a divergence-form operator with bounded measurable ellipticity, a drift in a Kato class, and forcing in a space with $3/p + 2/q < 2$. Here:

- ξ is a constrained system, not a scalar;
- the weight r^2 in $\mathrm{div}(r^2\nabla\xi)$ has ellipticity ratio $\sup_E r^2/\Omega^2$, uncontrolled without $\|\omega\|_{L^\infty}$;
- the drift is not $O(1/\Omega)$ and is not known to be in L^2;
- the one term that *is* in $L^2_tL^2_x$, namely $P_\xi(S\xi)$, satisfies $3/2 + 2/2 = 5/2 > 2$, so it is below the inhomogeneous Hölder threshold;
- even a successful De Giorgi–Nash–Moser argument produces some Hölder exponent $\alpha > 0$ depending on ellipticity, not necessarily $\alpha \ge 1/2$. Exponents $\beta < 1/2$ do not close the stretching estimate [2].

Parabolic Campanato for spatial Hölder 1/2 on cylinders of radius ρ (time-scale ρ^2) scales as $\rho^{-(3+2+1)}\int_{Q_\rho}|\xi - \bar\xi|^2 = \rho^{-6}\int_{Q_\rho}$, not as the elliptic ρ^{-4} integral. The restriction $\rho \le \Omega/2$ is dimensionally meaningless: ρ is a length and Ω is a vorticity threshold.

What remains true. On $\{r > 0\}$ the equation is (28), with the projection and the quadratic term. The stretching force $P_\xi(S\xi)$ lies in $L^2_tL^2_x$ globally by Calderón–Zygmund $2 \to 2$, without $\|\omega\|_{L^\infty}$. Definition 7.1 still requires a uniform Campanato bound (40) on $E(t)$. The argument above does not produce it.

References

[1] H. Beirão da Veiga and L. C. Berselli, *On the regularizing effect of the vorticity direction in incompressible viscous flows*, Differential Integral Equations **15** (2002), 345–356.

[2] H. Beirão da Veiga, *Open problems concerning the Hölder continuity of the direction of vorticity for the Navier–Stokes equations*, arXiv:1604.08083.

[3] P. Constantin and C. Fefferman, *Direction of vorticity and the problem of global regularity for the Navier–Stokes equations*, Indiana Univ. Math. J. **42** (1993), 775–789.

[4] C. Fefferman, *Existence and smoothness of the Navier–Stokes equation*, Clay Mathematics Institute Millennium Problems, 2000.

[5] J. T. Beale, T. Kato, and A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Comm. Math. Phys. **94** (1984), 61–66.

[6] T. Kato, *Strong L^p-solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions*, Math. Z. **187** (1984), 471–480.

[7] J. Serrin, *On the interior regularity of weak solutions of the Navier–Stokes equations*, Arch. Rational Mech. Anal. **9** (1962), 187–195.